

ECE 461 - HW05

1)

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \left\{ \begin{array}{l} \text{I}_4 \\ \text{P} \end{array} \right. \begin{array}{l} \rightarrow b_1 \\ \rightarrow b_2 \\ \rightarrow b_3 \\ \rightarrow b_4 \\ \rightarrow p_1 = b_1 \oplus b_2 \oplus b_3 \\ \rightarrow p_2 = b_2 \oplus b_3 \oplus b_4 \\ \rightarrow p_3 = b_1 \oplus b_2 \oplus b_4 \end{array}$$

$$1.a) \quad \vec{c} = [0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1]^T$$

Parity check matrix is $H = [P \ I_3] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

s.t. $HG = 0_{3 \times 4}$

syndrome : $H\vec{c} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

For a valid code word $\vec{c} \in \ker(H)$ but we see that $H\vec{c} \neq \vec{0}$, meaning this is NOT a codeword } codeword must be in the null-space of H or $H\vec{c} = 0$

* Alternatively, we could've checked parities:

For $b_1 = 0, b_2 = 0, b_3 = 0, b_4 = 1$, we must have

$$p_1 = 0 \oplus 0 \oplus 0 = 0, \quad p_2 = 0 \oplus 0 \oplus 1 = 1, \quad p_3 = 0 \oplus 0 \oplus 1 = 1$$

↳ This doesn't quite satisfy the parity part of our codeword $\vec{p} = [1 \ 1 \ 1]^T$
Thus, codeword is NOT valid

1.b) We've found that $H\vec{c} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

and that $H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$
5th column

$\Rightarrow$ error is in the 5th bit of the codeword $\vec{c}$.

Valid codeword is $\boxed{[0001011]^T}$

1.c) Let us try changing 1st bit to 1

$$\left. \begin{array}{l} b_1 = 1 \\ b_2 = 0 \\ b_3 = 0 \\ b_4 = 1 \end{array} \right\} \begin{array}{l} p_1 = 1 \\ p_2 = 1 \\ p_3 = 0 \end{array} \Rightarrow \text{If we change } p_3 \text{ to 0 as well,} \\ \text{it becomes a valid codeword}$$

Generated valid codeword = $\boxed{[1001110]^T}$

Received codeword w/ 2 errors = $[0001111]^T$

Let us try changing 3rd bit to 1

$$\left. \begin{array}{l} b_1 = 0 \\ b_2 = 0 \\ b_3 = 1 \\ b_4 = 1 \end{array} \right\} \begin{array}{l} p_1 = 1 \\ p_2 = 0 \\ p_3 = 1 \end{array} \Rightarrow \text{If we change } p_2 \text{ to 0 as well,} \\ \text{it becomes a valid codeword}$$

Generated valid codeword = $\boxed{[0011101]^T}$

Received codeword w/ 2 errors = $[0001111]^T$

* If we change b_2 or b_4 , it changes 2 parity bits which makes 3 errors in total.

↳ BUT if we flip b_2 and b_4 together

$$\left. \begin{array}{l} b_1 = 0 \\ b_2 = 1 \\ b_3 = 0 \\ b_4 = 0 \end{array} \right\} \begin{array}{l} p_1 = 1 \\ p_2 = 1 \\ p_3 = 1 \end{array} \Rightarrow \begin{array}{l} \text{Generated valid codeword} = [0100111]^T \\ \text{Received codeword w/ 2 errors} = [0001111]^T \end{array}$$

1.d) * If all of info bits were 1, then all parity bits will be 1 making it a valid codeword.

$$\begin{array}{l} \text{Generated valid codeword} \Rightarrow [1111111]^T \\ \text{Received codeword w/ 3 errors} \Rightarrow [0001111]^T \end{array}$$

* If we flip $b_2 = 1$, then

$$\left. \begin{array}{l} b_1 = 0 \\ b_2 = 1 \\ b_3 = 0 \\ b_4 = 1 \end{array} \right\} \begin{array}{l} p_1 = 1 \\ p_2 = 0 \\ p_3 = 0 \end{array} \Rightarrow \begin{array}{l} \text{Generated valid codeword} \Rightarrow [0101100]^T \\ \text{Received code w/ 3 errors} \Rightarrow [0001111]^T \end{array}$$

→ There are more instances of valid codewords w/ 3 errors such as [0000]

1.e) We see that minimum Hamming distance is 3 since flipping one of (b_1, b_3, b_4) changes 2 parity bits, (changing b_2 flips all parity bit), making 3 bit flips in total

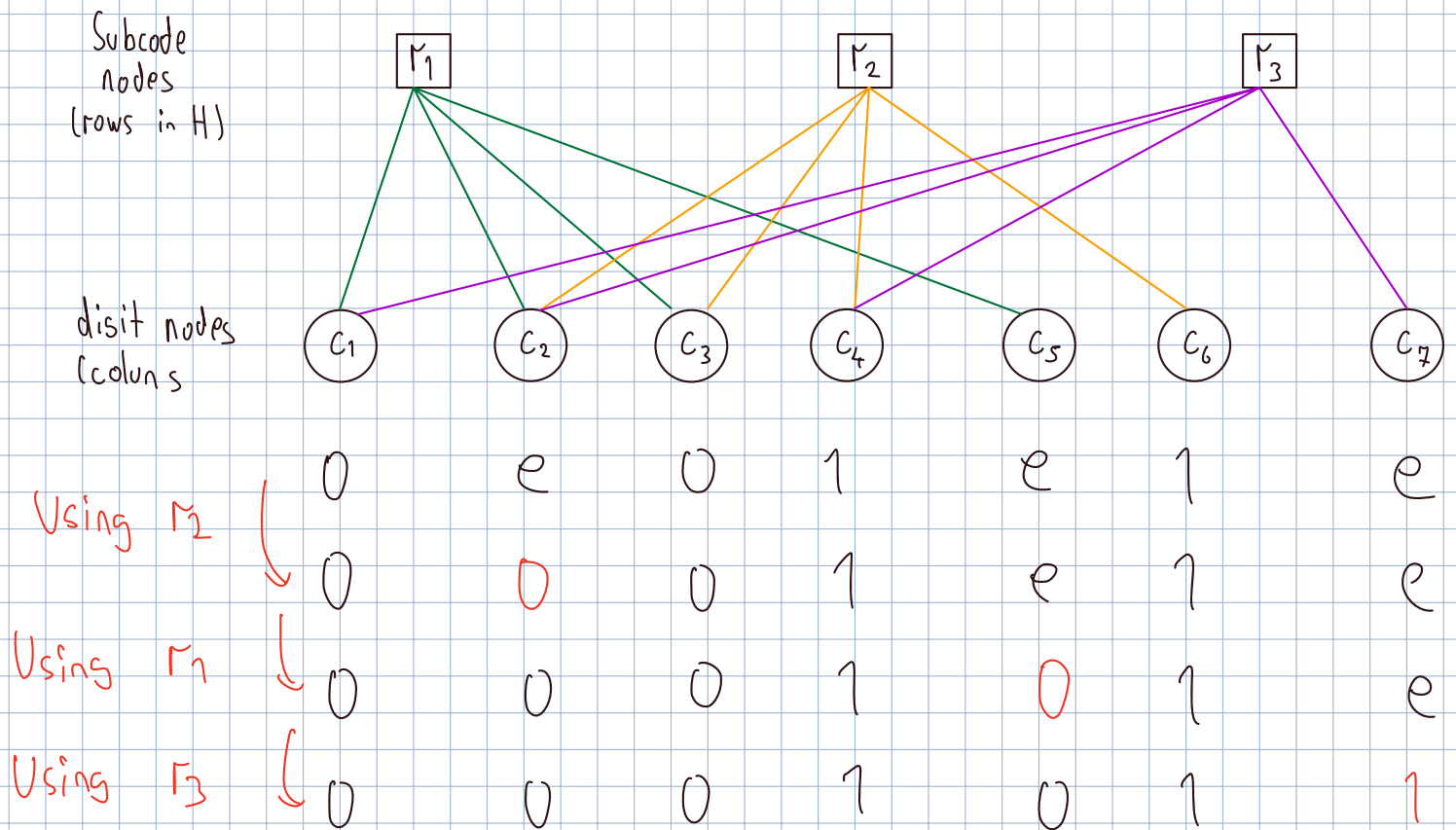
$$d_{\min} = \min_{k \neq r} d(c_k, c_r) = 3 = 2l + 1 \Rightarrow l = 1$$

↳ Thus, one can correct at most 1 bit at a time. In another words correction capability of $(7,4)$ Hamming code is 1 bit.

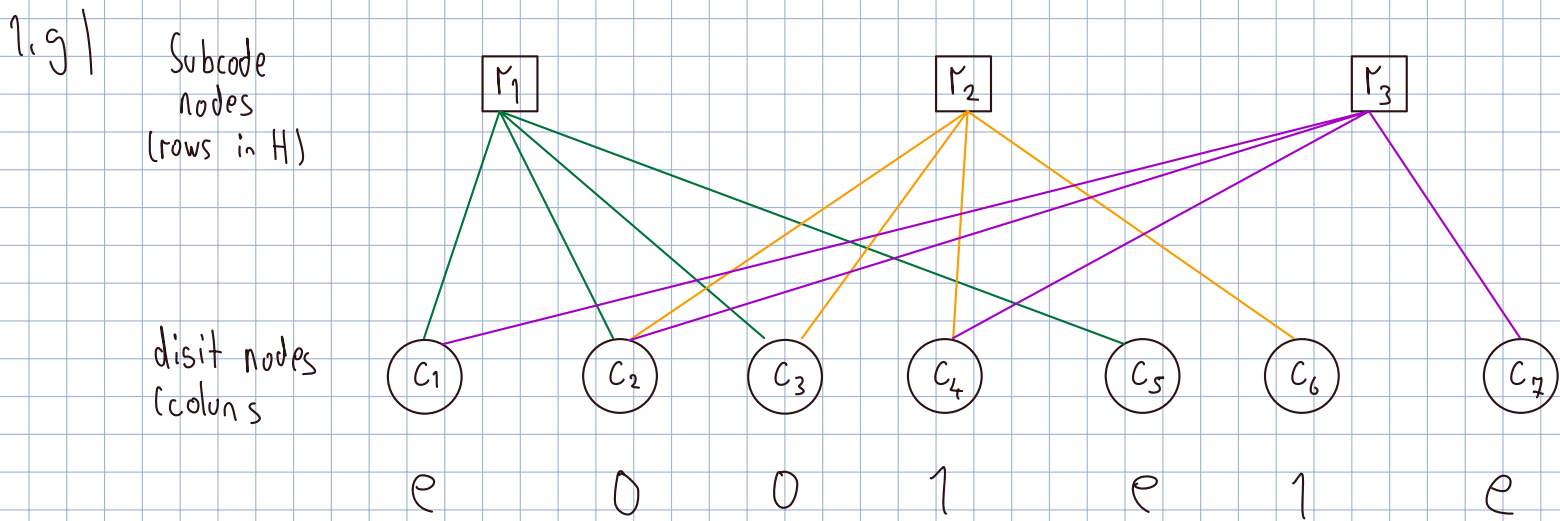
Therefore, codewords w/ 2 or 3 errors can not be corrected since there are more than 1 possible codeword.

However; if there is only 1 error, valid codeword is unique,

f) Using Message Passing Algorithm with $H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$



* We've used the fact that parities of each row must be 0 which is equivalent to $H\vec{c} = \vec{0}$,



We must have $\rightarrow c_1 \oplus c_5 = 0$ using r_1
 $\rightarrow c_1 \oplus c_7 = 1$ using r_3

$\hookrightarrow c_1 = 1, c_5 = 0, c_7 = 0$ and $c_1 = 0, c_2 = 0, c_7 = 1$ are both possible and form a valid codeword.

* We can't uniquely recover the original codeword.

This is because r_2 doesn't give us any additional info, We have 2 equations (from r_1, r_3) but 3 unknowns making the system underdetermined

2.a) We can calculate the expected number of total symbols transmitted after M transmissions as:

$$N_{\text{total}}(M) = \underbrace{n}_{1^{\text{st}}} + \underbrace{pn}_{2^{\text{nd}}} + \underbrace{p^2n}_{3^{\text{rd}} \text{ transmission}} + p^3n + \dots = n \sum_{n=0}^{M-1} p^n = n \frac{1-p^{M-1}}{1-p}$$

In each transmission $(1-p)$ of the symbols will be received without erasure. Expected value of total # of symbols w/o erasures is
At k -th transmission $p^{k-1}n$ symbols are sent and $1-p$ of those are without erasures.

$$N_{\text{we}}(M) = (1-p)n + (1-p)pn + (1-p)p^2n + (1-p)p^3n + \dots \\ = (1-p)n \sum_{n=0}^M p^n = (1-p)n \frac{1-p^{M-1}}{1-p} = n(1-p^{M-1})$$

Clearly for $N_{\text{we}}(M) = n$, we need $M \rightarrow \infty$.

$$\text{Then } N_{\text{total}} = \lim_{M \rightarrow \infty} n \frac{1-p^{M-1}}{1-p} = \boxed{\frac{n}{1-p}}$$

2.b) Each symbol encodes 1 bit in BEC, so rate is

$$R = \frac{\text{number of symbols sent}}{\text{average number of transmissions}}$$

* We've found that we need $N_{\text{total}} = \frac{n}{1-p}$ transmissions for sending n symbols.

$$R = \frac{n}{\frac{n}{1-p}} = \boxed{1-p}$$

$\nearrow$ As $p \rightarrow 0$: perfect binary channel every bit received correctly
 $\searrow$ As $p \rightarrow 1$: channel completely useless unreliable transmission

3.a) From average energy constraint, we find

$$E = \frac{\text{total energy}}{\# \text{ of channels}} = \frac{E_T}{M} \Rightarrow E_T = ME$$

Also since rate (R) is defined to be:

$$R = \frac{\text{total \# of bits}}{\# \text{ of channel}} = \frac{B}{M} \Rightarrow B = MR$$

Then energy per bit is $E_b = \frac{E_T}{B} = \frac{ME}{MR} = \frac{E}{R}$,

* Note that B here is the total number of bits transmitted using whole M channels.

3.b) If R is very close to C with high reliability

$$R \approx C = \frac{1}{2} \log(1 + \text{SNR})$$

$$E_b \approx \frac{E}{R} = \frac{\text{SNR} \sigma^2}{\frac{1}{2} \log(1 + \text{SNR})} \Rightarrow \boxed{\frac{E_b}{\sigma^2} = \frac{2 \text{SNR}}{\log(1 + \text{SNR})}}$$

↳ Since linear term on numerator grows faster than log term on denom. ; this is an increasing func. of SNR

$$3.c) \lim_{\text{SNR} \rightarrow 0} \frac{2 \text{SNR}}{\log_2(1 + \text{SNR})} = \lim_{\text{SNR} \rightarrow 0} \frac{2 \text{SNR}}{\frac{\ln(1 + \text{SNR})}{\ln 2}} = 2 \ln 2 \lim_{\text{SNR} \rightarrow 0} \frac{\text{SNR}}{\ln(1 + \text{SNR})}$$

Applying L'Hospital Rule and taking derivative of both num. and denom.

$$= 2 \ln 2 \lim_{\text{SNR} \rightarrow 0} \frac{1}{\frac{1}{1 + \text{SNR}}} = 2 \ln 2 \approx 1.4, \Rightarrow \text{Even if the SNR is low it is possible to operate near capacity if systems } E_b \text{ satisfies } E_b / \sigma^2 > 2 \ln 2$$

If the system is operating with high reliability, R being very close to upper bound C ; $SNR \gg 1$. This yields

$$R \approx \frac{1}{2} \log_2 SNR = \# \text{ bits/channel} \quad \text{when } SNR \gg 1$$

(high-SNR regime)